

2(a)(i)	$\text{area} = ut + \frac{1}{2}(v - u)t$ or $\text{area} = vt - \frac{1}{2}(v - u)t$ or $\text{area} = \frac{1}{2}(u + v)t$	A1
2(a)(ii)	displacement	A1
2(b)(i)	$u = 15 \sin 60^\circ (= 13 \text{ m s}^{-1})$	C1
	$t = 15 \sin 60^\circ / 9.81$	C1
	$= 1.3 \text{ s}$	A1
2(b)(ii)	the force in the horizontal direction is zero	B1
2(b)(iii)	(velocity =) $15 \cos 60^\circ = 7.5 \text{ (m s}^{-1}\text{)}$ or (velocity =) $15 \sin 30^\circ = 7.5 \text{ (m s}^{-1}\text{)}$	A1
2(c)(i)	$p = mv$ or 0.40×7.5 or 0.40×4.3	C1
	$\Delta p = 0.40 (7.5 + 4.3)$ $= 4.7 \text{ kg m s}^{-1}$	A1
2(c)(ii)	force = $4.7 / 0.12$ or $0.40 \times [(7.5 + 4.3) / 0.12]$ $= 39 \text{ N}$	A1